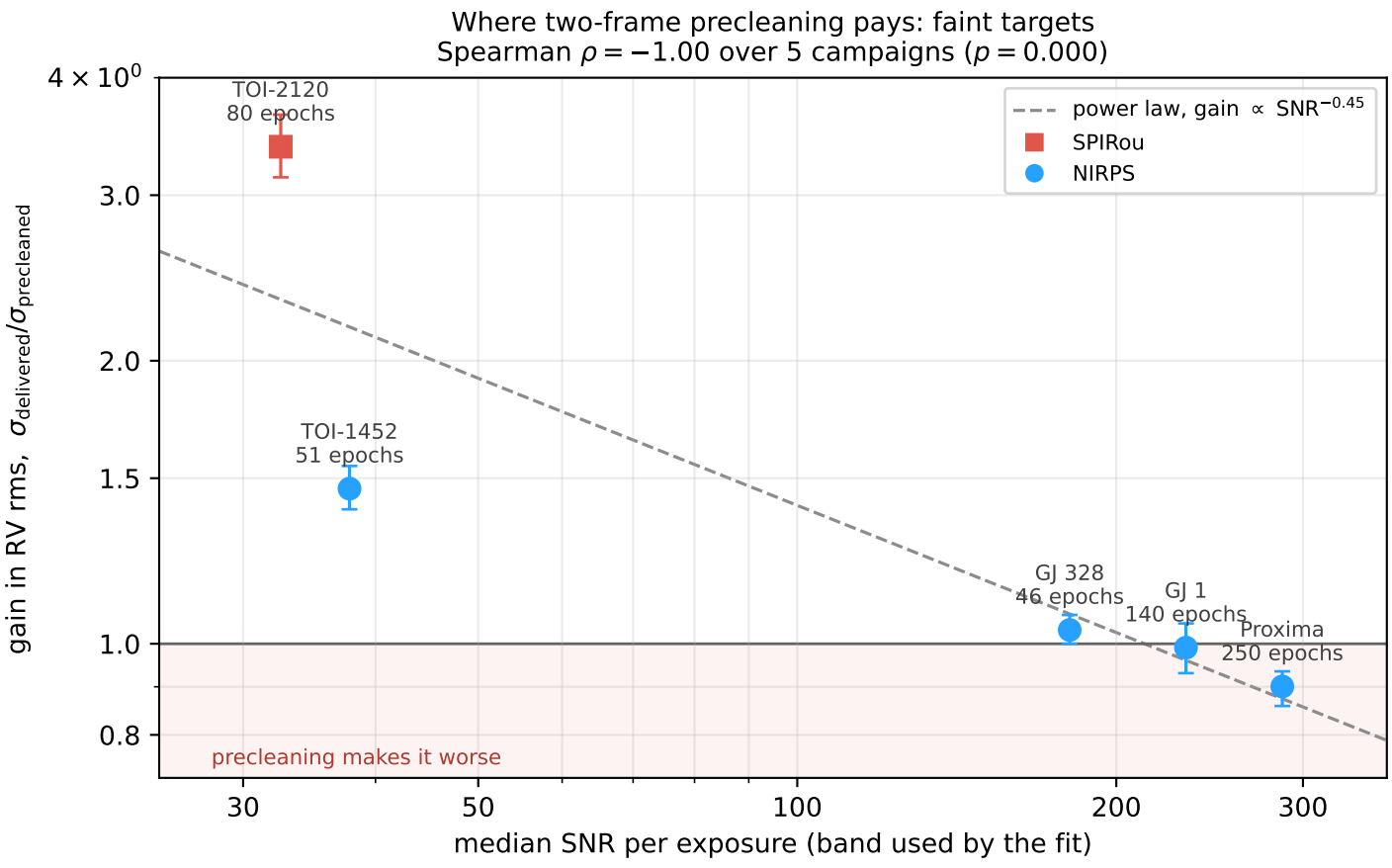
Where two-frame precleaning pays: faint targets
Spearman $\rho = -1.00$ over 5 campaigns ($p = 0.000$)
power law, gain $\propto$ SNR$^{-0.45}$
SPIRou
NIRPS
TOI-2120
80 epochs
TOI-1452
51 epochs
GJ 328
46 epochs
GJ 1
140 epochs
Proxima
250 epochs
precleaning makes it worse
gain in RV rms, $\sigma_{\mathrm{delivered}}/\sigma_{\mathrm{precleaned}}$
median SNR per exposure (band used by the fit)
4×10^0
3.0
2.0
1.5
1.0
0.8
30
50
100
200
300